MAJOR PROJECT

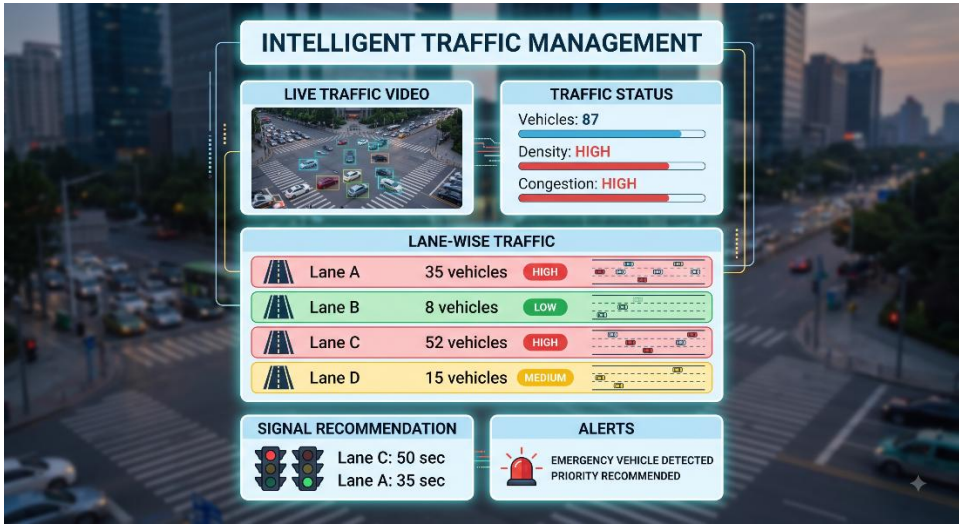
PROJECT TITLE: **Intelligent Traffic Management System Using Computer Vision**

1. **Introduction:**

- Most city traffic lights run on fixed timers. They ignore actual traffic, which is why you often sit at a red light on a packed road while empty lanes get green lights.
- Computer vision solves this by scanning live camera feeds. It counts cars and tracks lane density automatically so decisions match real road conditions.
- This software tracks cars from video, calculates lane congestion, and adjusts light timings on the fly. It also spots emergency vehicles like ambulances and gives them immediate priority.
- We are building the prototype with a central dashboard. You can track live car counts, check congestion levels, and view timing recommendations in one place.

2. **Problem Definition:**

- Traditional traffic signals run on pre-set timers. They ignore real-time changes on the road. A packed lane often gets the exact same green light time as an empty one, which stacks up traffic and wastes road space.
- Manual traffic checks need people watching screens full-time. Humans get tired, miss details, and cannot track density across multiple intersections at once.
- Cities need a system that watches roads on its own. Computer vision processes live camera feeds to help managers adjust light timings based on actual traffic.
- The proposed system addresses this problem by processing traffic video, detecting and counting vehicles, tracking vehicles across lanes, estimating traffic density and generating traffic signal timing recommendations according to the observed traffic conditions.



3. Literature Survey:

No.	Paper	Year	Method	Dataset / Environment	Main Work	Results	Limitation	Relevance
1	You Only Look Once: Unified, Real-Time Object Detection	2016	YOLO	PASCAL VOC, COCO	Proposed YOLO as a single-stage real-time object detection method.	Base YOLO operated at about 45 FPS; Fast YOLO reached about 155 FPS.	Early YOLO had difficulty with small objects and localization.	Foundation for real-time vehicle detection.
2	Simple Online and Realtime Tracking	2016	Kalman Filter + Hungarian Algorithm	MOT benchmarks	Proposed a simple and fast method for online multi-object tracking.	Tracker operated at about 260 Hz with competitive tracking performance.	Sensitive to detection quality and difficult occlusions.	Useful for tracking vehicles across video frames.
3	Simple Online and Realtime Tracking with a Deep Association Metric	2017	DeepSORT	MOT16 / Re-ID data	Added deep appearance features to SORT for better object association.	Reduced identity switches by about 45% while maintaining high frame rates.	Additional appearance-model computation increases complexity.	Useful for maintaining vehicle IDs during tracking.

No.	Paper	Year	Method	Dataset / Environment	Main Work	Results	Limitation	Relevance
4	YOLOv4: Optimal Speed and Accuracy of Object Detection	2020	YOLOv4	MS COCO	Improved YOLO architecture and training methods for real-time detection.	Reported 43.5% AP and about 65 FPS on Tesla V100.	High-performance operation requires GPU resources.	Suitable for real-time traffic-object detection.
5	YOLOv7: Trainable Bag-of-Freebies Sets New State-of-the-Art for Real-Time Object Detectors	2022	YOLOv7	MS COCO	Improved real-time object detection through architectural and training techniques.	Reported 56.8% AP for the evaluated real-time detector range.	GPU resources are required for high-speed deployment.	Candidate model for vehicle detection.
6	UA-DETRAC: A New Benchmark and Protocol for Multi-Object Detection and Tracking	2020	Detection + Multi-Object Tracking	UA-DETRAC	Developed a traffic-specific benchmark for vehicle detection and tracking.	100 traffic videos, over 140,000 frames and about 1.21 million vehicle bounding boxes.	Limited geographic and environmental diversity.	Excellent dataset and benchmark for traffic detection/tracking.
7	BDD100K: A Diverse Driving Dataset for Heterogeneous Multitask Learning	2020	Deep Learning / Multi-task Vision	BDD100K	Developed a large-scale driving dataset covering multiple computer-vision tasks.	100K driving videos with geographic, weather and environmental diversity.	Dash-camera perspective differs from fixed intersection cameras.	Useful for training and evaluating vision models.
8	Vehicle Detection in Intelligent Transportation Systems and Its Applications Under Varying Environments: A Review	2018	Computer Vision Review	Multiple traffic datasets	Reviewed vehicle detection methods and their applications in intelligent transportation systems.	Identified major detection approaches and environmental challenges.	Review rather than a single implemented system.	Provides theoretical foundation for vehicle detection in ITS.
9	Vehicle Detection in Intelligent Transport System Under a Hazy Environment: A Survey	2020	Computer Vision Review	Multiple traffic studies	Reviewed vehicle detection under degraded environmental conditions.	Discussed detection, traffic density, queue length, speed and vehicle classification.	Mainly focuses on environmental degradation.	Useful for handling real-world traffic-camera conditions.

No.	Paper	Year	Method	Dataset / Environment	Main Work	Results	Limitation	Relevance
10	IntelliLight: A Reinforcement Learning Approach for Intelligent Traffic Light Control	2018	Deep Reinforcement Learning	Real traffic data from surveillance cameras	Developed an intelligent traffic-light controller that dynamically responds to traffic conditions.	Demonstrated improved adaptive traffic-light control using real traffic data.	RL training and deployment are more complex than fixed-rule systems.	Directly relevant to adaptive signal control.
11	Deep Reinforcement Learning for Traffic Light Control in Vehicular Networks	2018	DQN + CNN / Deep RL	Traffic simulation	Used deep reinforcement learning to dynamically control traffic signals.	Demonstrated improved traffic-control efficiency in simulated scenarios.	Mainly evaluated in simulation.	Supports AI-based adaptive signal timing.
12	Learning Phase Competition for Traffic Signal Control	2019	Reinforcement Learning / FRAP	Traffic simulation	Modeled competition between traffic-signal phases according to traffic demand.	Improved convergence and generalization under different traffic conditions.	Requires RL training and simulation environments.	Useful for selecting which traffic direction receives priority.
13	PressLight: Learning Max Pressure Control to Coordinate Traffic Signals in Arterial Network	2019	Reinforcement Learning + Max Pressure	Traffic simulation	Combined transportation theory with reinforcement learning for signal control.	Improved traffic performance compared with conventional and several learning methods.	More suited to coordinated signal networks.	Useful for developing traffic-density-based signal decisions.
14	CoLight: Learning Network-Level Cooperation for Traffic Signal Control	2019	Graph Attention + Reinforcement Learning	Large-scale traffic simulations	Enabled multiple traffic signals to cooperate when making decisions.	Demonstrated strong performance on large traffic networks.	More complex than a single-intersection system.	Useful for future expansion to multiple intersections.
15	CityFlow: A Multi-Agent Reinforcement Learning Environment for Large Scale City Traffic Scenario	2019	Multi-Agent RL Simulator	Synthetic + real-world traffic data	Developed a high-performance traffic simulator for intelligent traffic-signal research.	Reported more than 20× the speed of SUMO in the described comparison.	Simulation results may not directly represent real-world deployment.	Useful for testing signal-control algorithms without hardware.

No.	Paper	Year	Method	Dataset / Environment	Main Work	Results	Limitation	Relevance
16	Recent Advances in Reinforcement Learning for Traffic Signal Control: A Survey of Models and Evaluation	2021	Systematic Review	Multiple traffic-control studies	Reviewed RL approaches, state representations, rewards and evaluation methods for traffic signals.	Compared major reinforcement-learning approaches used for signal control.	Conclusions depend on previously published experimental results.	Provides background for AI-based adaptive signals.
17	Reinforcement Learning for Traffic Signal Control: Comparison with Commercial Systems	2021	DQN + Actor-Critic	Simulated road networks	Compared reinforcement-learning traffic controllers with established traffic-signal systems.	RL methods achieved lower delay in evaluated scenarios.	Primarily simulation-based evaluation.	Useful for comparing intelligent and conventional traffic control.
18	Reinforcement Learning in Urban Network Traffic Signal Control: A Systematic Literature Review	2022	Systematic Literature Review	160 peer-reviewed studies	Reviewed reinforcement-learning methods for urban network traffic signals.	Covered research from 30 countries and identified major research directions.	Focuses mainly on signal control rather than computer-vision detection.	Provides broad research context for intelligent traffic management.
19	Deep Reinforcement Learning for Traffic Light Control in Intelligent Transportation Systems	2023	DQN + DDPG	Single-intersection and grid simulations	Studied adaptive traffic-light control using deep reinforcement learning.	Demonstrated adaptive signal policies in simulated traffic environments.	Results are primarily simulation-based.	Supports development of the traffic decision module.
20	Intelligent Traffic Signal Control Based on Reinforcement Learning: A Survey	2026	RL / Deep RL Survey	Multiple traffic-control studies	Reviews reinforcement-learning approaches for traffic-signal phase selection and timing.	Summarizes current RL approaches and deployment challenges.	Real-world scalability and deployment remain challenging.	Provides recent research context for intelligent traffic management.

➢ Overall Limitations of Existing Research

- **Real-world deployment**: Many systems are tested mainly in simulations.
- **Camera dependency:** Poor lighting, weather and camera angles reduce accuracy.
- **Vehicle occlusion**: Overlapping vehicles cause detection and tracking errors.
- **Small vehicles:** Distant vehicles are difficult to detect accurately.
- **High computation:** Advanced models may require powerful GPUs.
- **Dataset limitations:** Existing datasets may not represent Indian traffic.
- **Limited integration**: Detection, tracking and signal control are often developed separately.
- **Generalisation**: Models may perform poorly in new locations or conditions.
- **Signal-control complexity:** Reinforcement learning requires extensive training.
- **Emergency vehicles**: Emergency priority is not consistently included.
- **Scalability**: Systems may struggle when expanded to multiple intersections.
- **Mixed traffic:** Indian roads include cars, bikes, buses, auto-rickshaws and irregular traffic behaviour.

4. Research Gap:

- Existing software handles vehicle tracking, car counts, and traffic light timing on paper. The problem is that researchers usually build these pieces in isolation instead of combining them into one smooth pipeline.

- Most models rely on heavy server hardware or run inside sterile computer simulations. They struggle when you put them on real street corners with limited processing power.

- Traffic in India adds another layer of chaos. Drivers ignore lane markers, while autorickshaws, motorcycles, and cows share the asphalt with buses. Standard datasets trained on clean Western highways fail here.

- Cities need an integrated setup that processes live camera footage on standard computers. It needs to spot mixed traffic types, measure lane congestion, and update signal timers right away.

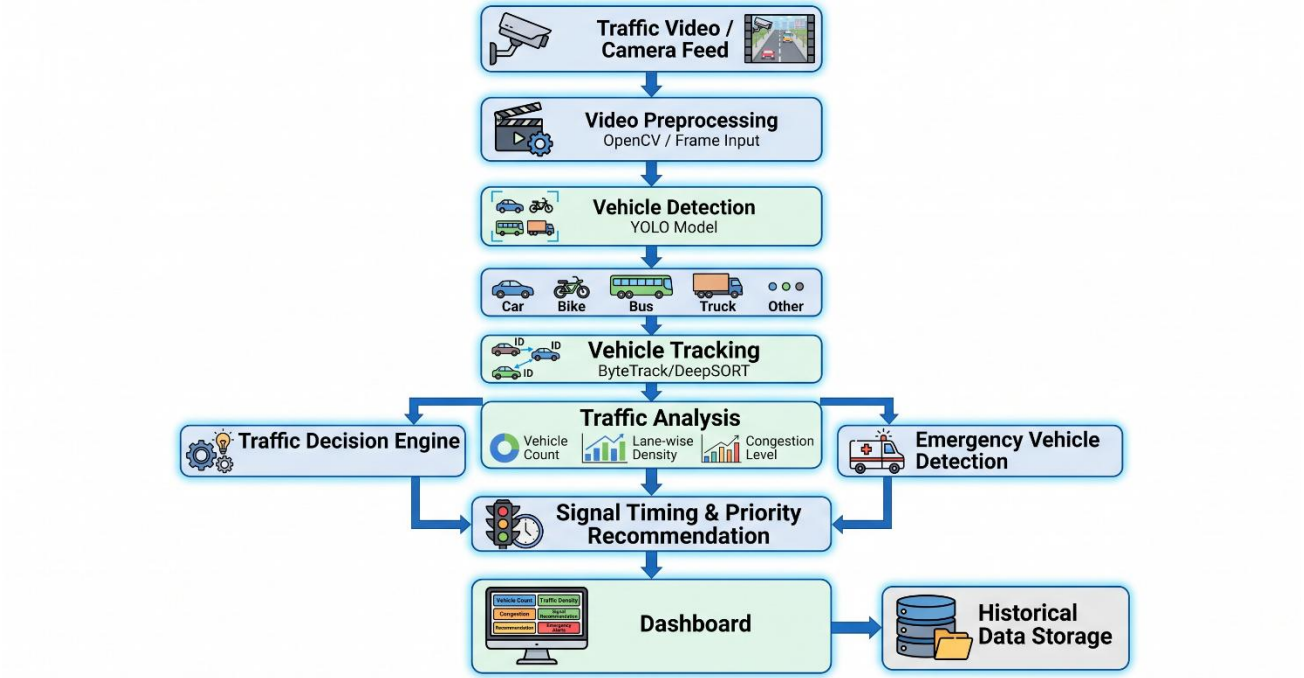
5. Objectives:

- **Primary Objective**
  To develop a computer-vision-based intelligent traffic management system that analyses traffic video in real time and assists in making adaptive traffic-control decisions.

- **Specific Objectives**

1. **Vehicle Detection**
   Detect cars, buses, trucks, motorcycles and other vehicles from traffic-camera footage.
2. **Vehicle Tracking**
   Track detected vehicles across consecutive video frames.
3. **Vehicle Counting**
   Calculate the number of vehicles passing through different lanes.
4. **Traffic Density Estimation**
   Determine traffic density for each lane or direction.
5. **Congestion Detection**
   Identify low, medium and high traffic congestion.
6. **Speed Estimation**
   Estimate vehicle speed where camera calibration permits.
7. **Adaptive Signal Recommendation**
   Recommend signal timings based on current traffic conditions.
8. **Emergency Vehicle Priority**
   Detect emergency vehicles and provide priority recommendations.
9. **Traffic Dashboard**
   Display vehicle counts, density, congestion and system decisions visually.
10. **Performance Evaluation**
    Compare the proposed approach using metrics such as detection accuracy, FPS, counting accuracy and traffic waiting time.

6. Scope of the Project:



Scope includes:

- Traffic-video analysis
- Vehicle detection
- Vehicle classification
- Vehicle tracking
- Lane-wise counting
- Traffic-density estimation
- Congestion detection
- Speed estimation
- Adaptive signal recommendations
- Emergency vehicle detection
- Visualization/dashboard
- Performance evaluation

Not <u>included</u> initially

- Physical traffic-light hardware
- Direct control of real traffic signals
- Installation of roadside cameras
- City-wide deployment
- Legal enforcement decisions

“The proposed system will initially be developed as a software prototype and evaluated using recorded or publicly available traffic videos.”

7. Novelty

<u>Proposed Novel Features:</u>

1. **Integrated Computer Vision Pipeline**
   Detection, tracking, counting and traffic analysis are combined in one system.
2. **Lane-wise Traffic Analysis**
   Instead of only counting total vehicles, the system analyses traffic for individual lanes/directions.
3. **Dynamic Signal Recommendation**
   Signal recommendations are generated according to current traffic density rather than fixed timings.
4. **Emergency Vehicle Priority**
   Emergency vehicles can be detected and assigned higher priority.
5. **Traffic Dashboard**
   The system provides a visual interface showing:
   - Vehicle count
   - Vehicle types
   - Traffic density
   - Congestion level
   - Signal recommendation
6. **Software-Based Implementation**
   The prototype can operate using traffic video without requiring dedicated traffic sensors or physical hardware.

7. Feasibility Study:

I. Technical Feasibility

The project is technically feasible because open-source technologies are available.

Component	Proposed Technology
Programming	Python
Computer Vision	OpenCV
Object Detection	YOLO
Object Tracking	DeepSORT / ByteTrack
Data Processing	NumPy, Pandas
Visualization	Streamlit / Flask
Dataset	UA-DETRAC, BDD100K, public traffic videos
Traffic Simulation	SUMO / CityFlow
Development	VS Code / PyCharm

II. Economic Feasibility

- You can build the system using free, open-source software and public datasets.
- The first prototype does not need custom traffic lights or specialised physical hardware.
- This setup keeps your upfront development costs near zero.

III. Operational Feasibility

- Traffic management staff run the software through a central dashboard.
- The screen displays live road conditions and suggests signal timings based on the processed video.

IV. Schedule Feasibility

Phase	Work
Phase I	Problem definition + literature survey
Phase II	Dataset collection + preprocessing
Phase III	Vehicle detection
Phase IV	Tracking + counting
Phase V	Density + congestion analysis
Phase VI	Signal recommendation
Phase VII	Dashboard
Phase VIII	Testing + evaluation
Phase IX	Documentation + final presentation

8. Multidisciplinary / Transdisciplinary Aspects:

Computer Science / AI

- Machine learning
- Deep learning
- Computer vision
- Object detection
- Object tracking

Transportation Engineering

- Traffic density
- Traffic flow
- Congestion
- Signal timing
- Vehicle movement

Data Science

- Dataset processing
- Statistical analysis

- Performance evaluation
- Visualization

Software Engineering

- System architecture
- Backend
- Dashboard
- Testing
- Deployment

Possible transdisciplinary aspect

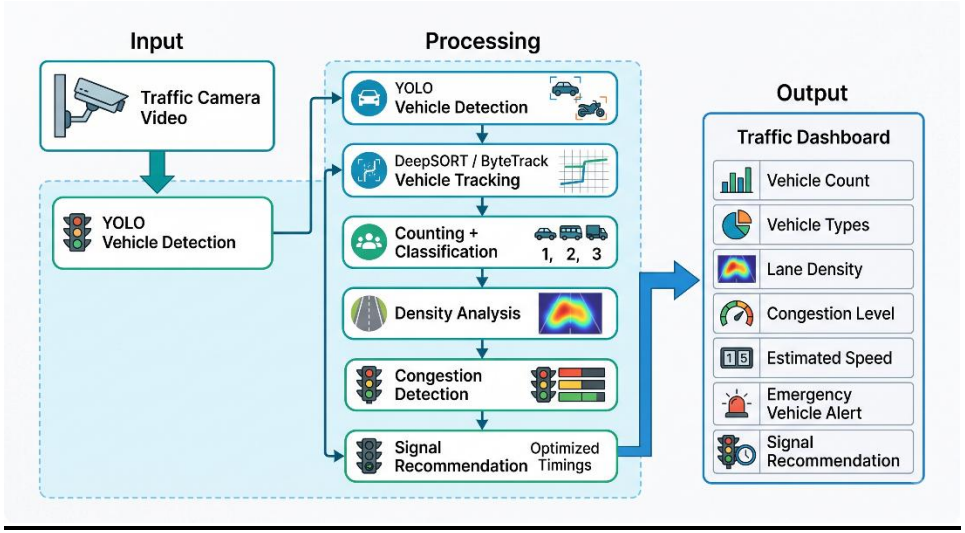
 We combine computer vision and artificial intelligence with traffic control rules to help cities make smarter road decisions.

 The system analyses live video feeds to count cars and track movement across different lanes.

 Traffic teams get clear, accurate data on current road conditions instead of relying on guesswork or fixed timers.

 This automated approach reduces bottlenecks by helping signals adapt to real-time traffic flow.

9. Product Development Scope:



10. Components and Bill of Materials

Component	Requirement	Purpose
Computer/Laptop	Existing system	Development and testing
Python	Free	Programming
OpenCV	Free/Open Source	Video processing
YOLO	Open Source	Vehicle detection
DeepSORT/ByteTrack	Open Source	Vehicle tracking
NumPy	Free/Open Source	Numerical processing
Pandas	Free/Open Source	Data processing
Streamlit/Flask	Free/Open Source	Dashboard
Public datasets	Free	Training/testing
SUMO/CityFlow	Free/Open Source	Traffic simulation/testing

Hardware requirement

No dedicated hardware is required for the initial prototype. The system can be developed and tested using an existing computer and recorded traffic videos.

11. Proposed System Architecture:

